

农友天气 WeatherNext — User Guide

A weather forecasting tool for Malaysian agriculture

What is WeatherNext?

WeatherNext is a multi-lingual (Chinese / English / Bahasa Malaysia / Tamil / Burmese) weather forecast app for farms across Malaysia. It provides 14-day forecasts for 60+ pre-loaded Malaysian locations and lets you add your own farm spots.

The app is a Progressive Web App (PWA) — install it once and it works like a native app on your phone or computer, even offline after the first visit.

Live URL: <https://stanleywoosweeleong.github.io/weathernext/>

Installing the App

On iPhone / iPad

1. Open Safari and go to the live URL above
2. Tap the ****Share**** button (square with arrow pointing up) at the bottom of the screen
3. Scroll down and tap ****Add to Home Screen****
4. Tap ****Add**** in the top right corner
5. The WeatherNext icon will appear on your home screen — tap it to open

On Android Phone

1. Open Chrome and go to the live URL
2. Tap the ****menu**** (three dots, top right)
3. Tap ****Install app**** or ****Add to Home screen****
4. Confirm the installation

On Windows Desktop

1. Open Chrome or Edge and go to the live URL
2. Click the ****install icon**** in the address bar (looks like a screen with a down arrow)
3. Click ****Install****
4. WeatherNext will open as a standalone desktop app

****Tip for Windows users:**** The app is designed for mobile, so it may appear small/large on a desktop screen. Press ****Ctrl**** + ****Plus (+)**** a few times to zoom to a comfortable reading size. Chrome remembers this setting.

Setting Up Your Personal Google API Key (Required for AI Features)

WeatherNext includes AI-powered farming features (智能农事简报, 病虫害风险分析, AI 策划与咨询) that need a free Google Gemini API key to work. Each user provides their own key — it's free, takes about 5 minutes, and is stored only on your device.

Why do you need your own key?

The AI features call Google's Gemini AI model. Google requires every user to have their own key so they can give everyone a generous free quota (around 1,500 requests per day) without one person's usage affecting others.

Step 1 — Get your free key from Google

1. Open this URL in your browser: ****https://aistudio.google.com/apikey****
2. Sign in with any Google account (your Gmail account works fine)
3. Click the blue ****"Create API key"**** button
4. If asked, choose ****"Create API key in new project"**** (the default)
5. After a few seconds, Google shows you a long string starting with ``Alza...`` — that's your key
6. Click the ****copy**** icon next to the key to copy it to your clipboard

7. ****Save this key somewhere safe**** — you can also get it again later from the same page, but it's easier to keep a copy

Step 2 — Add the key to WeatherNext

1. Open the WeatherNext app
2. Tap any of the AI buttons (purple **** 🌟 智能农事简报 ****, green **** 🌿 病虫害风险分析 ****, or blue **** 🌟 AI 策划与咨询 ****)
3. If no key is set, a popup appears asking for your ****API Key****
4. ****Paste**** your key (long-press and tap Paste on phone, or Ctrl+V on desktop)
5. Tap ****Save Key****
6. The AI feature will now work — and the key is remembered for future visits

Important notes about your API key

- ****The key is stored only on your own device.**** It never gets sent to anyone else.
- ****Each device needs its own key entry**** — if you install the app on both your phone and laptop, you'll need to paste your key on each one.
- ****The key is free**** — Google's free tier is more than enough for normal personal use.
- ****If you lose your key****, just visit <https://aistudio.google.com/apikey> again to see it or create a new one.
- ****Don't share your key with others**** — anyone with your key could use up your daily quota.

Troubleshooting AI features

If an AI feature fails with an error:

- ****"401" or "API key invalid"**** — your key was typed/pasted incorrectly. Re-enter it.
- ****"429" or "Quota exceeded"**** — you've used up the daily free quota. Try again tomorrow.
- ****No popup asking for key**** — the app already has a key. To replace it: tap the AI button → the popup might show your current key, you can clear and re-enter.

Adding Your Own Farm Locations

The app comes with 60+ pre-loaded locations across Malaysia, but you can add your own farm spots for accurate forecasts at your exact location.

Step 1 — Find your farm's coordinates

You need the **latitude** and **longitude** of your farm. Here are the easiest ways to get them:

Method A: Google Maps (recommended)

1. Open **Google Maps** on your phone or computer
2. Search for your farm address, OR navigate to your farm on the map
3. **Long-press** (on phone) or **right-click** (on computer) on the exact spot
4. A small popup or red pin appears
5. On phone: the coordinates appear at the top (like `4.2105, 101.9758`)
6. On computer: right-click → click the coordinates to copy them
7. The format is **latitude, longitude** — the first number is latitude (around 1 to 7 for Malaysia), the second is longitude (around 99 to 120 for Malaysia)

Method B: I'm at my farm right now

If you're physically at the farm:

1. Open Google Maps
2. Tap the **blue dot** that shows your current location
3. A card at the bottom shows your coordinates
4. Copy both numbers

Method C: WhatsApp Location Share

If someone shared your farm's location via WhatsApp:

1. Open the location they shared
2. Tap **"View on map"** or similar
3. Google Maps opens — follow Method A from there

Step 2 — Add the location in WeatherNext

1. In WeatherNext, tap the ** (magnifying glass)** in the top right of the header
2. Tap **"Add your own farm spot / 添加您的地点"** at the top
3. Fill in the form:
 - **English name** (or leave blank if you only want Chinese)
 - **Chinese name** (or leave blank if you only want English)
 - **Latitude** — the first number (e.g., `4.2105`)
 - **Longitude** — the second number (e.g., `101.9758`)
 - You can also tap **"Upload Photo of Compass/Map"** to use an AI feature that extracts coordinates from a photo
4. Tap **Save**
5. Your new location appears in the list, automatically pinned as a favorite

Tips for adding locations

- **Name it whatever you want** — "My Durian Farm", "Bukit Tinggi Block A", "Mike's Lot 12" — whatever helps you find it
- **You can leave one language blank** — if you only speak Chinese or only English, just fill that one
- **Coordinates must be in Malaysia-ish** — the app warns you if coordinates seem outside Malaysia (latitude 0.8–7.8, longitude 99–120), but you can still save them
- **Accuracy matters for weather** — try to pick the exact farm location, not just the nearest town, especially for areas with mountains or coastlines
- **You can add many locations** — there's no hard limit

Managing your locations

- **To favorite/unfavorite a location** — tap the ❤️ heart icon next to its name
- **To rename a location** — tap the **Edit / 改名** option (look for the edit icon)
- **To delete a custom location** — go to the search modal, scroll down to "Deleted Items", or use the delete option in the location row (built-in locations cannot be deleted, only custom ones)

Using the Forecast Details

Tap any location to open the **14-day detailed forecast modal**. You'll see:

| Section | What It Shows |

|-----|-----|

| **WVET Crop Stress** | Wet/Vegetation/Evapotranspiration risk (color-coded: green = low, yellow = moderate, orange = high, red = severe) |

| **Temp / Dew Pt / DeltaT** | Max temperature, dew point, and DeltaT (8AM) for spraying safety |

| **Wind / Direction / Gusts** | Wind speed in km/h, direction arrows, gust values |

| **Cloud (%) & CB Risk** | Cloud cover by altitude, plus CB Risk indicator when conditions suggest convective storms |

| **Sky Tint (AM/PM)** | Visual representation of expected sky color/clarity morning and afternoon |

| **Rain Probability** | Chance of rain by day |

| **Rainfall & ET0** | Expected rainfall and reference evapotranspiration in mm |

Switching between AI Model and ECMWF

In the top right of any location's details, tap the **AI Model** or **ECMWF** pill to switch forecasts:

- **AI Model** — Google's latest AI-based forecast (sometimes more accurate for short-term)
- **ECMWF** — European weather model (industry standard for 5-14 day forecasts)

Try both and compare for your area.

Switching crop context

Tap the orange **榴莲/果树** pill to switch crop context. The AI features will customize their advice based on which crop you select.

AI Features Explained

These three buttons appear at the top of each location's forecast modal:

🌟 智能农事简报 (Smart Farming Briefing)

A 2-3 paragraph summary of the next 7 days written specifically for your crop. Highlights spraying windows, irrigation needs, and weather risks.

🦟 病虫害风险分析 (Pest & Disease Risk Analysis)

Analyzes the forecast for conditions that favor common pests and diseases (high humidity, prolonged leaf wetness, temperature ranges). Tells you what to watch for and when.

✨ AI 策划与咨询 (AI Planning & Consulting)

Open-ended chat where you can ask any farming question and get advice contextualized with the actual weather forecast for your location.

Other Useful Features

- **Pull-down to refresh** — pull down from the top of the main list to refresh weather data
- **Pin favorites** — tap the heart icon to pin frequently-used locations at the top
- **Tap-to-zoom in forecast modal** — pinch to zoom in/out, double-tap to reset
- **Download forecast** — the download icon in the forecast modal saves the chart as an image you can share via WhatsApp

Troubleshooting

App seems stuck or showing old data

- Pull down to refresh
- Or look for the green **RELOAD** banner near the top — tap it to load the latest version

Can't see all 14 days

- The first time you open a forecast, scroll right to see all dates
- The last column (e.g. `28/5`) should be visible with margin on both sides

Forecast says "--/--" or doesn't load

- Check your internet connection
- The app uses Open-Meteo weather API — if their service is down, forecasts won't load

Lost my favorites after deleting the app

- Favorites are stored both locally on your device AND in cloud sync
- If you sign in (anonymous auth happens automatically), favorites should restore when you reinstall
- If they don't restore, just re-pin the locations you want

App not working at all

- Try opening the live URL in regular Safari/Chrome (not the installed PWA)
- If it works in the browser but not the installed app, delete and reinstall the app
- If it doesn't work in the browser either, there may be a deployment issue — contact the developer

Quick Reference Card

| Task | How |

|-----|-----|

| Install on iPhone | Safari → Share → Add to Home Screen |

| Install on Windows | Chrome/Edge → Install icon in address bar |

| Get AI key | <https://aistudio.google.com/apikey> → Create API key |

| Add AI key | Tap any AI button → paste key → Save |

| Find coordinates | Google Maps → long-press location → copy lat/long |

| Add custom location |  → Add your own farm spot → fill form → Save |

| Switch language | Tap EN / BM / 中文 / தமிழ் / မြန်မာ in header or modal |

| Pin location | Tap  next to location name |

| View forecast | Tap any location name |

| Reset zoom in forecast | Double-tap the chart |

| Download forecast image | Download icon (↓) in forecast modal footer |

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App developed by Stanley Woo Swee Leong. For technical issues, contact the developer.